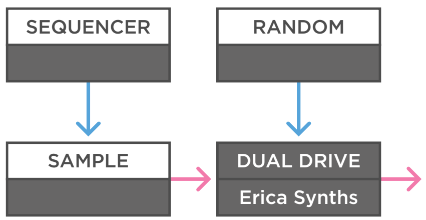
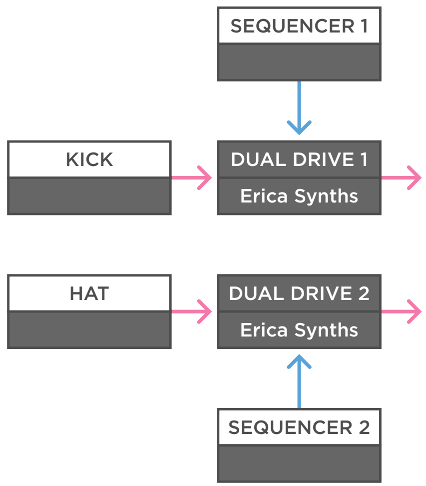
Erica Synths Dual Drive

Mono Patches

Audio
Control Voltage
Trigger / Gate



(a)



(b)

a - The Dual Drive works great to bring dynamics to mono drum or synth patches. For example, if you use a single sample player with single drum hits and use a simple sequencer to sequence those hits. You can use a smooth random voltage, somewhat related to the tempo of the drums, to modulate the drive input. This way you give a simple drum pattern some random dynamics over time and make it less static.

b - You can also take a steady kick drum and hi-hat, and feed each of the sounds to one side of the Dual Drive. Then use two sequences with different step lengths to modulate the drive or gain, for example, to create subtle motion in the kick, and a bit more drastic motion in the hats.

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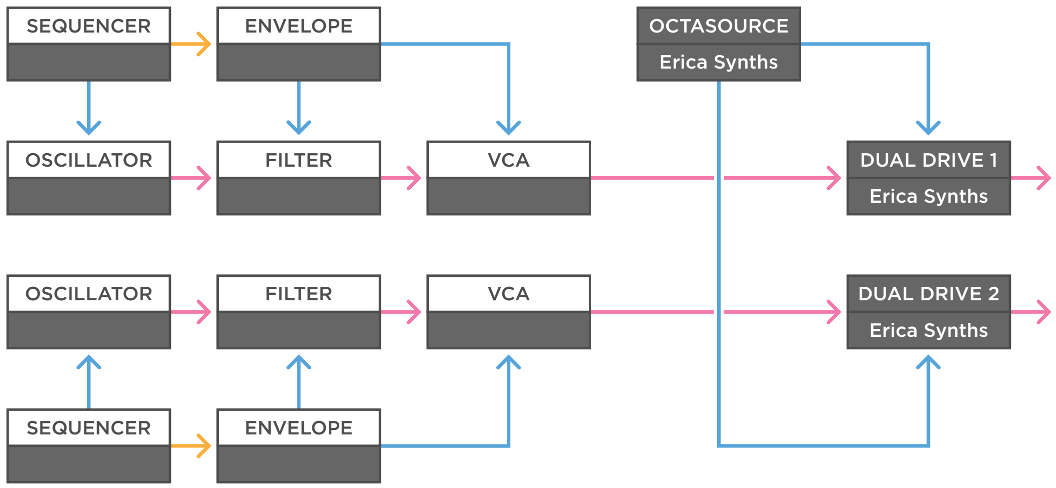

(a)

a - The Dual drive also works great on synth lines, especially more dynamic ones. Here are two simple synth voices creating sequenced melodies. Each voice is sent to one side of the Dual Drive. Then you can send an LFO to both drive CV inputs.

One regular and one with inversed polarity, for example, with two sine waves from the Black Octasource. This creates a nice tension between the two synth voices. When one overdrives, the other becomes more gentle, and the other way around.

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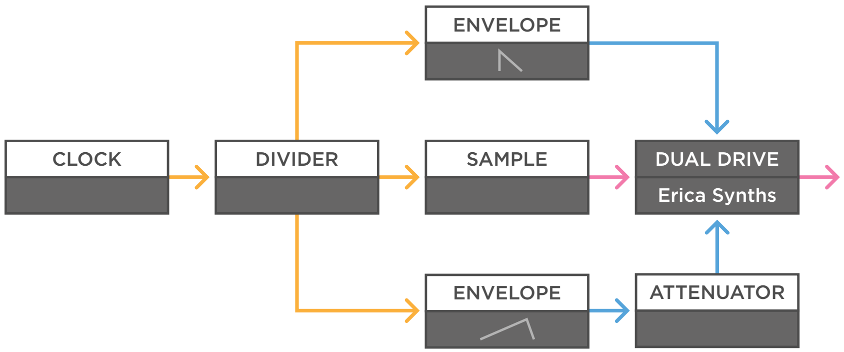
MONOTRAIL
TECH TALK



(a)

a - You can use both CV inputs on the Dual Drive to create more complex dynamics. For example, let’s use a clock to drive a clock divider and use a division to trigger a simple steady synth sample. The sample is sent to the Dual Drive with both the gain and drive set pretty low. Then you can use a faster clock division to trigger a short envelope and feed that to the drive CV.

And in addition, you can use a slower division to trigger a longer envelope and send that to the gain input. However, because there is only one attenuator, you have to decide which signal you want to have the most effect – in this case that would be the fast envelope to the gain. And then use an attenuator to control the signal with the more subtle influence.

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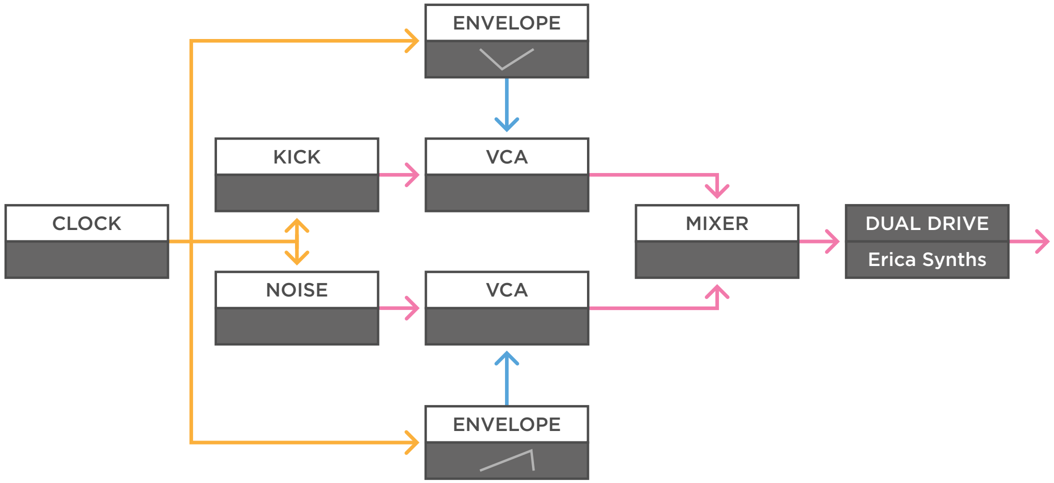
MONOTRAIL TECH TALK



(a)

a - If you're into more grungy techno, the Dual Drive is also a great way to sculpt your kick. For example, if you take a kick drum and a sample player playing a gritty noise sample, you can send both those sounds to independent VCAs. When you use a steady clock to trigger both samples, you can multiply that clock signal, and send it to two envelopes.

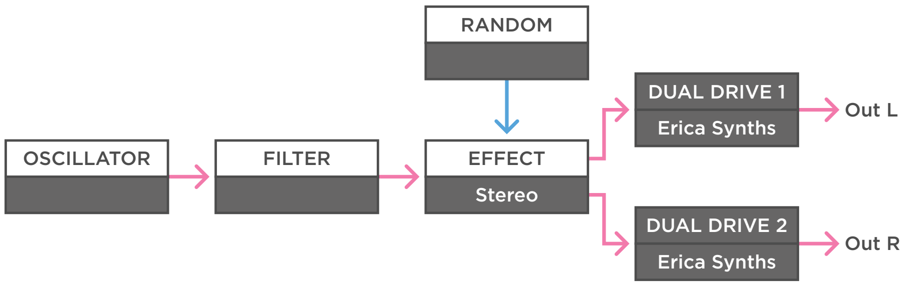
One inverted, with the attack shaping the length of the initial kick, and the decay set to bring the end of a longer kick back up. The second envelope is used on the VCA of the gritty sample with a longer rising envelope to increase the volume of the grit after the kick hits. Then, when you mix those signals and send it to the Dual Drive, it can really glue those sounds together to make one massive sound.

Erica Synths Dual Drive

Stereo Patches

MONOTRAIL
TECH TALK

Audio
Control Voltage
Trigger / Gate



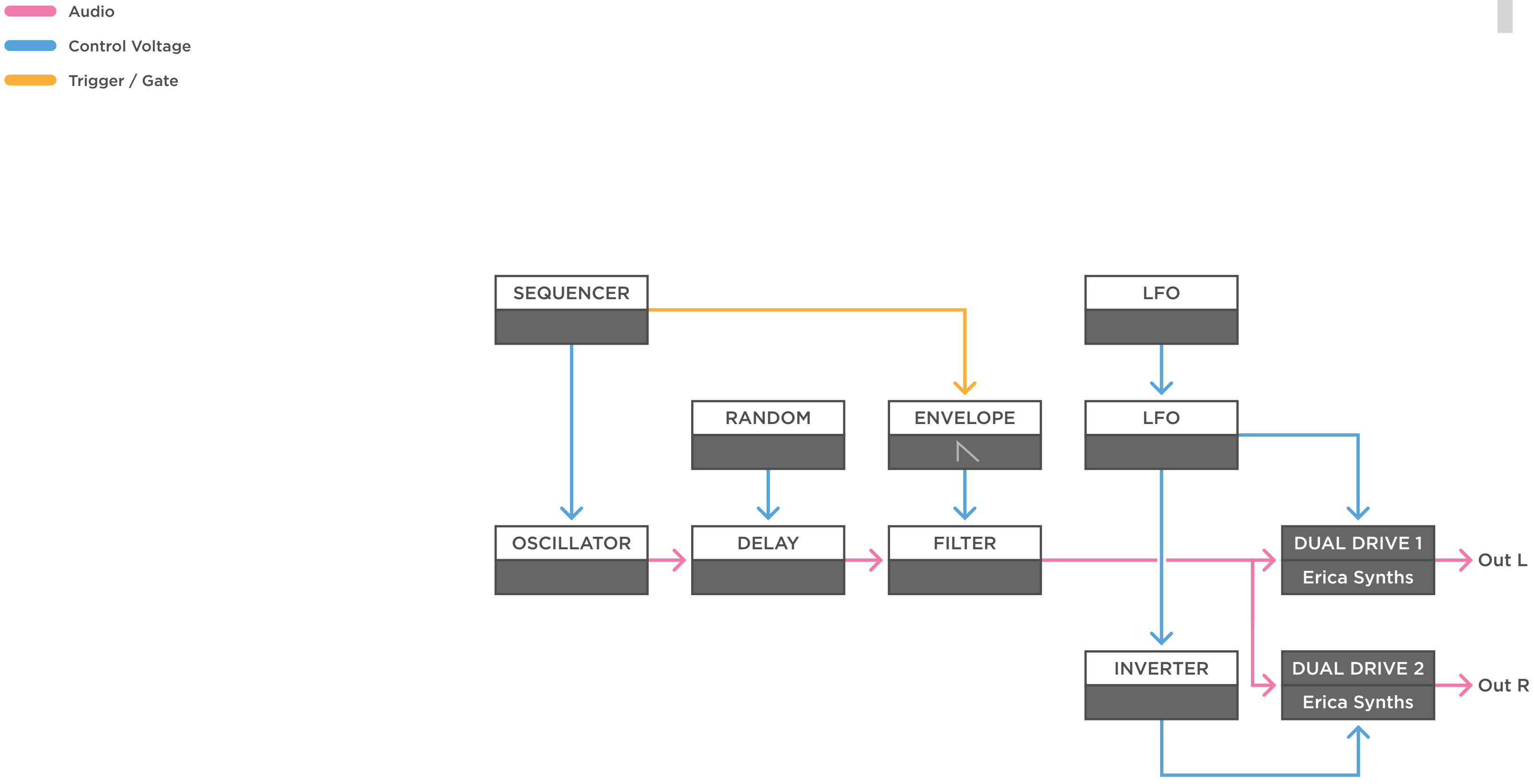
(a)

a - Because the Dual Drive offers two identical circuits, it works great on stereo modular patches. For example, in this patch, an oscillator and filter are creating a low drone. Then the signal is sent through a stereo effect module, to create a nice stereo texture.

The effect is modulated with a bit of random voltage, and then the left and right output of the effect are sent through the two sides of the Dual Drive, to overdrive the stereo sound.

Erica Synths Dual Drive

Stereo Patches



(a)

a - But you can also use the Dual Drive to make stereo patches, which works great on drums and synth voices. Here, a simple voice is made with a single oscillator, delay, and filter. That signal is multiplied to the two independent sides of the Dual Drive to create a stereo signal. A slow random voltage is sent to the delay to create a synth line with a lot of harmonics, and a short attack decay envelope is used on the filter for a pulsating effect.

A sequencer is used to create a simple melody and trigger the envelope. Finally, a regular and an inverted sine wave are used to modulate each of the drive CV inputs. And to make it more dynamic, another slow sine wave LFO is used to modulate the frequency of the LFO creating the stereo signal.

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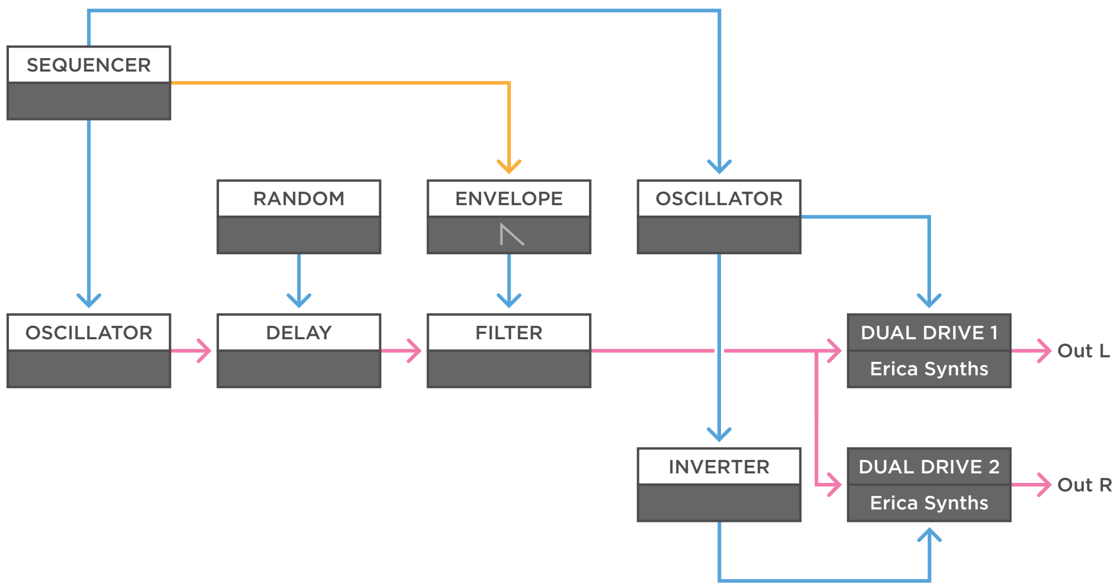

(a)

a - You can experiment with this same patch, by for example using audio-rate modulation instead of the two inverted sine waves. Use another tuned oscillator but remember to make sure to multiply and invert one of the signals before modulating the Dual Drive. Working with audio-rate modulation creates a more steady fat stereo sound.

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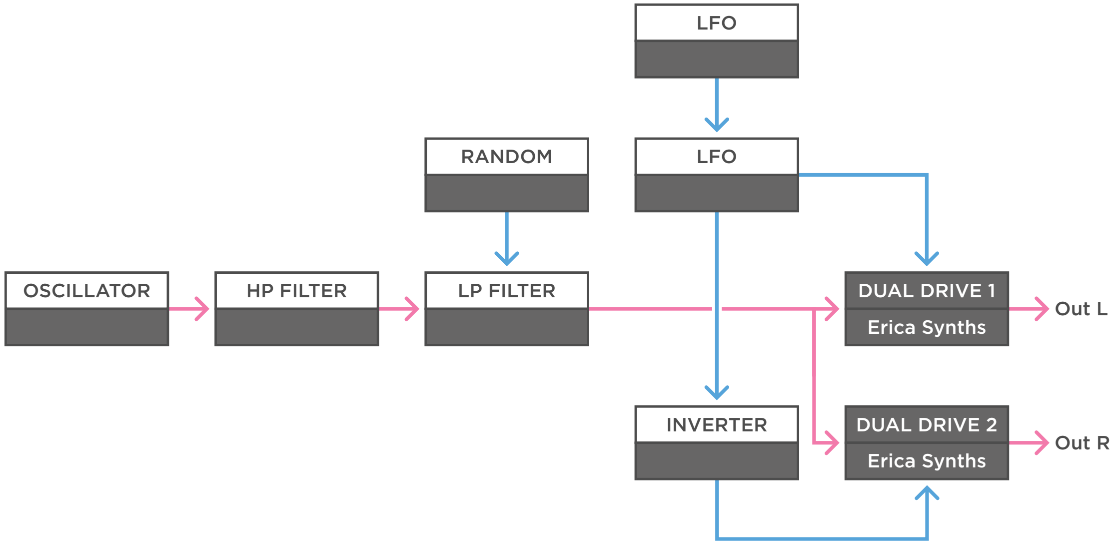

(a)

a - Using the Dual Drive to create stereo sounds easily leads to great results. Here is another example with a single oscillator though a resonant high pass and low pass filter. This time with some slow random modulation to the low pass filter, and two square waves modulating both drive CV inputs. Again with a bit of modulation to the square wave LFO to have the tempo drift over time.

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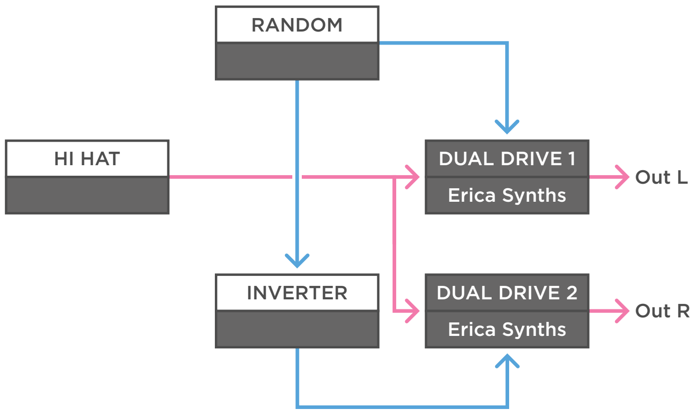
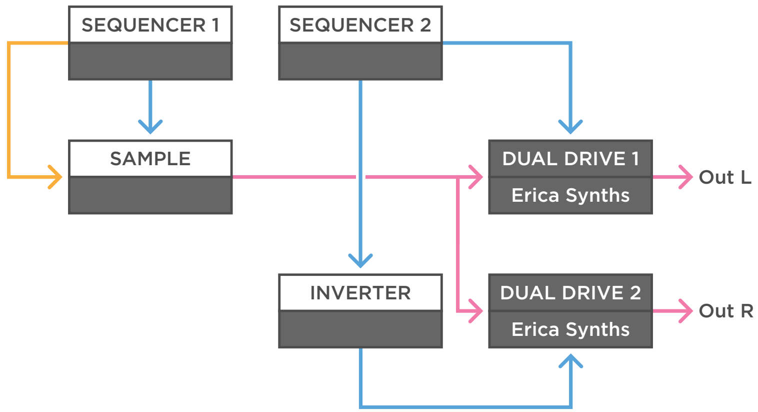


(a)



(b)

a - Of course this stereo concept also works on drums. For example, you can take a steady hi-hat pattern, but multiply the signal, feed those to two sides of the Dual Drive, and use them as your left and right input. Then take a smooth random voltage and send it to one of the CV inputs. Lastly, send a copy through an inverter, to the CV input of the second circuit.

b - Or you can use the sample player to create a drum pattern again. One sequence is used to trigger the drums and select samples. A second sequence is used to the Drive CV input, and again, a copy of that signal is sent through an inverter to the second Drive CV input. By adjusting the attenuators on the Dual Drive, you can create interesting stereo overdrive. Here, an 8 step sequence is used to sequence the drums, and a 7 step sequence to create a stereo field on the Dual Drive.